Joint probability distribution

Joint probability distribution for X,Y is a probability distribution that gives the probability that each X,Y falls in any particular range or discrete set of values specified for that variable. It is also called bivariate distribution.

Discrete case:

$$P(x,y)=\sum_i\sum_j P(X=x_i, Y=y_j)$$

Joint density function:

Continuous case:

$$f(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dxdy$$

Marginal distribution of X and Y

Discrete case:

$$P(x)=\sum_y P(x,y); \qquad P(y)=\sum_x P(x,y)$$

Continuous case:

$$f(x)=\int_{-\infty}^{\infty} f(x,y)\,dy; \qquad f(y)=\int_{-\infty}^{\infty} f(x,y)\,dx$$

Conditional distribution:

Discrete case:

Conditional probability function of X, given $Y=y_j$

$$P\left(X=x_i / Y=y_j\right)=\frac{P(X=x_i \,\&\, Y=y_j)}{P(Y=y_j)}=\frac{P_{ij}}{P_j}$$

Conditional probability function of Y, given $X=x_i$ **Dr.K.KAVITHA**

$$P\left(\frac{Y = y_j}{X = x_i}\right) = \frac{P(X = x_i \,\&\, Y = y_j)}{P(X = x_i)} = \frac{P_{ij}}{P_i}$$

Continuous case:

Conditional distribution of the random variable Y given that X=x is

$$f\left(\frac{y}{x}\right) = \frac{f(x, y)}{f(x)}, f(x) > 0$$

Conditional distribution of the random variable X given that Y=y is

$$f\left(\frac{x}{y}\right) = \frac{f(x, y)}{f(y)}, f(y) > 0$$

Independent Random variables

Discrete case:

The random variables X and Y are said to be independent random variables if $P_{ij} = P_i \times P_j$

Where P_{ij} is the joint probability function of (X,Y), P_i is the marginal probability function of X and P_j is the marginal probability function of Y.

Continuous case:

Two random variables X and Y are said to be independent if f(x,y)=f(x).f(y) where f(x,y) is the joint probability density function of (X,Y). f(x) is the marginal density function of X and f(y) is the marginal density function of Y.

Problems:

1. Given the following bivariate probability distribution. Compute the constant K.

X \ Y	1	2	3	4	5	6
2	3K	5K	K	K	2K	0
4	0	2K	3K	5K	K	K
6	K	2K	K	2K	K	2K

Also compute 1) $P(X \le 3, Y = 4)$ 2) $P(X \le 2)$ 3) $P(Y \le 4)$

4) $P(X < 3, Y \le 4)$ 5) $P(X + Y \le 4)$

Given that the distribution is a probability distribution

$$\therefore \sum_x \sum_y P(x, y) = 1$$

$$\Rightarrow 33k = 1$$

$$\Rightarrow k = \frac{1}{33}$$

Marginal distribution of X

X	1	2	3	4	5	6
P(x)	4/33	9/33	5/33	8/33	4/33	3/33

Marginal distribution of Y

Y	2	4	6
P(y)	12/33	12/33	9/33

1) $P(X \le 3, Y = 4) = P(X = 1, Y = 4) + P(X = 2, Y = 4) + P(X = 3, Y = 4)$

$$= 0 + 2/33 + 3/33 = 5/33$$

2) $P(X \le 2)$=P(X=1 or 2)=4/33+9/33=13/33

3) $P(Y \le 4)$ =P(Y=2 or 4)

=12/33+12/33=24/33

4) $P(X<3, Y\le 4) = P(X<3, Y=2) + P(X<3, Y=4)$

$$= P(X=1, Y=2) + P(X=2, Y=2) + P(X=1, Y=4) + P(X=2, Y=4)$$

$$= 3/33 + 5/33 + 0 + 2/33 = 10/33$$

5) $P(X+Y\le 4) = P(X=1, Y=2) + P(X=2, Y=2)$

$$=3/33+5/33=8/33$$

2. Given the following distributions of X and Y. Find the conditional distribution of X given Y=2.

X \ Y	1	2	3
0	1/15	2/15	1/15
1	3/15	2/15	1/15
2	2/15	1/15	2/15

X \ Y	1	2	3	P(x)
0	1/15	2/15	1/15	4/15
1	3/15	2/15	1/15	6/15
2	2/15	1/15	2/15	5/15
P(y)	6/15	5/15	4/15	

The conditional distribution of X given y=2

$$P\left(X=x_i \middle/ Y=2\right), x_i = 0,1,2$$

$$P\left(X=0 \middle/ Y=2\right) = \frac{P(X=0, Y=2)}{P(Y=2)} = \frac{2/15}{5/15} = 2/5$$

$$P\left(X=1 \middle/ Y=2\right) = \frac{P(X=1, Y=2)}{P(Y=2)} = \frac{2/15}{5/15} = 2/5$$

$$P\left(X=2/_{Y=2}\right)=\frac{P(X=2,Y=2)}{P(Y=2)}=\frac{1/15}{5/15}=1/5$$

3. Given the following joint density function

$$f(x)=\begin{cases}\frac{8}{k}xy, & 1\le x\le y\le 2\\ 0 & otherwise\end{cases}$$

1) Compute the value of k
2) Find the marginal densities
3) Find the conditional density function of X given Y.

Given that f(x,y) is a joint pdf.

$$\therefore \int_1^2\int_x^2 f(x,y)\,dydx = 1$$

$$\Rightarrow \int_1^2\int_x^2 \frac{8}{k}xydydx = 1$$

$$\Rightarrow \int_1^2\left[\frac{8}{k}x\frac{y^2}{2}\right]_x^2 dx = 1$$

$$\Rightarrow \frac{4}{k}\int_1^2 x(4-x^2 dx = 1$$

$$\Rightarrow K = 9$$

$$\therefore f(x)=\begin{cases}\frac{8}{9}xy, & 1\le x\le y\le 2\\ 0 & otherwise\end{cases}$$

Marginal density function

Marginal density function of X

$$f(x) = \int_x^2 f(x,y)dy$$

$$= \int_x^2 \frac{8}{9} xydy$$

$$= \left[\frac{8}{9} x \frac{y^2}{2} \right]_x^2 = \frac{4x}{9}(4 - x^2), \quad 1 \le x \le 2$$

Marginal density function of Y

$$f(y) = \int_1^y f(x,y)dx$$

$$= \int_1^y \frac{8}{9} xydx$$

$$= \left[\frac{8}{9} y \frac{x^2}{2} \right]_1^y = \frac{4y}{9}(y^2 - 1), \quad 1 \le y \le 2$$

Conditional density function of X given $Y = y$ is

$$f({}^{x}\!/\!_{y}) = \frac{f(x,y)}{f(y)} = \frac{\frac{8}{9}xy}{\frac{4y}{9}(y^2-1)} = \frac{2x}{y^2-1}, 1 \le x \le y \le 2$$

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4. Find the value of K if $f(x,y) = k(1-x)(1-y), \quad 0 < x, y < 1$ is a joint pdf.

Given that f(x,y) is the joint pdf.

$$\int_0^1 \int_0^1 f(x,y)\,dxdy = 1$$

$$\int_0^1 \int_0^1 k(1-x)(1-y)\,dxdy = 1$$

$$k\int_0^1 \int_0^1 (1-y-x+xy)\,dxdy = 1$$

$$K = 4$$

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